

# ANSWER SHEET — WRITING SCRIPT

Copy the left column by hand. The right column is for you only — do not write it on the sheet.

---

## Before you start

- Use four separate sheets of ruled paper. Write on one side only.
- Number each sheet at the bottom right: Page 1 of 4, Page 2 of 4, and so on.
- Write naturally. Do not print neatly — messy is the point.
- Cross out a word or two somewhere and rewrite it. Real scripts have corrections.
- Photograph or scan the four sheets in order and combine them into one PDF.
- The order below is deliberately scrambled. Follow it exactly or the ground truth file will not match.

## Header to write at the top of Page 1 only:

Name: Student 01 | Roll No: TEST-001 | Subject: Computer Science Fundamentals

## SHEET 1

---

**Q1.**

(b) Queue

**Q3.**

False. A primary key cannot contain NULL values because it must uniquely identify every row.

*Q3 comes before Q2 — out-of-order case number 1.*

**Q2.**

Random Access Memory.

**Q5.**

(b) Quick Sort

*Deliberately WRONG. Correct answer is (c) Merge Sort. Your grader should mark this incorrect and award 0.*

**Q6.**

Time complexity is a measure of the amount of time an algorithm takes to run, expressed as a function of the size of its input. Big-O notation is used because it describes the upper bound on the growth rate, which lets us compare algorithms without depending on the hardware.

**Q8.**

Normalization is the process of organising the data in a database so that redundancy is reduced and update anomalies are avoided. Two normal forms are First Normal Form (1NF) and Second Normal Form (2NF).

*Note that Q4 and Q7 have both been skipped on this sheet.*

## SHEET 2

### Q10.

A deadlock is a situation in which two or more processes are each waiting for a resource that is held by another process, so none of them can ever proceed. One necessary condition is mutual exclusion.

*Only ONE condition given where two were asked. Should earn partial marks (2 of 3), not full and not zero.*

### Q9.

TCP is connection oriented whereas UDP is connectionless. TCP guarantees delivery and preserves ordering, while UDP does neither. TCP is slower because of acknowledgements, so UDP is preferred for live streaming.

*Q9 written after Q10 — out-of-order case number 2.*

### Q11(a).

A hash table stores key-value pairs. A hash function converts the key into an index of an underlying array, and the value is placed at that index, which gives average constant time lookup.

*Sub-part labels must be extracted as two separate entries, not one Q11.*

### Q11(b).

A collision occurs when two different keys are hashed to the same index. One resolution technique is separate chaining, in which every index holds a linked list of all the entries that hash to it.

## SHEET 3

---

**Write only this one answer on Sheet 3, and stop exactly where the text stops — mid-sentence, at the bottom of the sheet.**

**Q12.**

Virtual memory is a memory management technique in which the system gives the illusion of a very large main memory by using secondary storage as an extension of it. In paging, the logical address space of a process is divided into fixed size blocks called pages, and physical memory is divided into frames of the same size. A page table maps every page number to a frame number. When a process refers to a page that is not present in main memory a page fault is raised, and the

*STOP HERE. Do not finish the sentence.  
This is the multi-page answer case — the  
answer continues on Sheet 4 and your  
highlighting must span both.*

## SHEET 4

**(no label — start writing straight away)**

operating system loads that page from the disk into a free frame. Two advantages are that a process larger than the available physical memory can still be executed, and that the degree of multiprogramming is increased.

*No question number at the top of the sheet. Your mapper cannot rely on labels alone here — it has to carry the previous question forward.*

**Q13.**

The TCP/IP model is a four layer networking model made up of the Link layer, the Internet layer, the Transport layer and the Application layer. Compared with the OSI model it has four layers instead of seven. The OSI Session, Presentation and Application layers are merged into the single TCP/IP Application layer, and the OSI Physical and Data Link layers are merged into the Link layer.

*This answers the OR alternative only. The first version of Q13 (the OSI one) must be reported as not attempted, NOT as a wrong answer.*

**Q14(b).**

45 divided by 2 gives 22 remainder 1. 22 divided by 2 gives 11 remainder 0. 11 divided by 2 gives 5 remainder 1. 5 divided by 2 gives 2 remainder 1. 2 divided by 2 gives 1 remainder 0. 1 divided by 2 gives 0 remainder 1. Reading the remainders upwards, 45 in binary is 101101.

*Sub-parts answered in reverse order — (b) before (a).*

**Q14(a).**

Recursion is a technique in which a function calls itself in order to solve a smaller instance of the same problem. For example factorial(n) is defined as n multiplied by factorial(n-1), with factorial(0) equal to 1.

**Q4.**

HyperText Transfer Protocol.

*Answered last, four sheets away from its printed position — out-of-order case number 3, and the hardest one.*

**Q15.**

Big Data refers to data sets that are so large or so complex that traditional data processing tools cannot handle them.

*There is no Q15 on the paper. This must be surfaced as an unmatched answer, not silently dropped and not forced onto Q14.*

**Rough Work**

101101

1024 512 256 128 64 32 16 8 4 2 1

*An unlabelled block. Either report it as unmatched or attach it to Q14(b) with low confidence — both are defensible, just be explicit about which you chose.*